

Student Name

March 31st, 2025.

Spring 2025. Physics 510.

Quantum Theory II.

Test # 2.

Show all your work step by step and write down your final answers explicitly. Plain final answers will not be graded. You are allowed to use your equation sheet. A4, one side, handwritten by you personally.

1. Consider a three-dimensional repulsive ($V_0 > 0$) square barrier of radius a .

$$V(r) = \begin{cases} V_0, & r < a \\ 0, & r \geq a \end{cases}$$

A particle of energy $E = \frac{\hbar^2 k^2}{2m} < V_0$ is incident on the well.

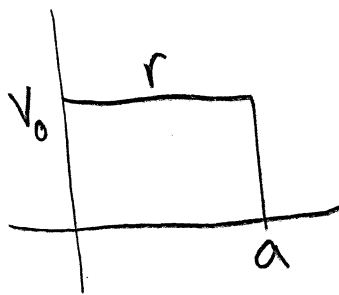
- Write down the Schrödinger equation for an s-wave ($l = 0$ mode).
- Solve it directly for both regions (it is possible because $l = 0$). Do not forget that the asymptotically free solution must contain a partial phase shift.
- Derive the phase shift, δ_0 , for the mode by applying necessary boundary conditions.
- How does the phase shift behave if $V_0 \rightarrow \infty$?
- Derive the total cross-section of the mode, σ_0 , in the limit of zero energy.

2. Calculate the scattering amplitude and differential cross-section in the Born approximation for the same potential

$$V(r) = \begin{cases} V_0, & r < a \\ 0, & r \geq a \end{cases}$$

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1.)



$$E = \frac{\hbar^2 k^2}{2m} < V_0$$

$$V_0 > 0$$

$$a) \quad \psi'' - V(r)\psi = E\psi$$

$$\psi'' - V(r)\psi = \frac{\hbar^2 k^2}{2m} \psi$$

$$b) \quad \text{for } r < a \quad \psi'' - V_0 \psi = \frac{\hbar^2 k^2}{2m} \psi$$

$$\frac{\psi''}{\psi} - \left(V_0 + \frac{\hbar^2 k^2}{2m} \right) = 0$$

$$\text{general solution is: } \psi = A e^{iqr} + B e^{-iqr}$$

$$A=0$$

$$\left(\frac{\hbar^2 k^2}{2m} \right) B e^{-iqr} \neq V_0 + \frac{\hbar^2 k^2}{2m} \quad l=0$$

$$R_{kl} = A_l j_l(kr) + B_l n_l(kr)$$

$$j_l(kr) \sim \frac{1}{kr} \sin(kr) \quad j'_l = \frac{1}{r} \cos(kr)$$

$$n_l(kr) \sim -\frac{1}{kr} \cos(kr) \quad n'_l = -\frac{k}{r} \sin(kr)$$

$$n''_l = \frac{k}{r} \cos(kr)$$

$$R_{kl}'' - V_0 R_{kl} = \frac{\hbar^2 k^2}{2m} R_{kl}$$

$$R_{kl}'' = -\frac{k}{r} \left(-A_l \sin(kr) - B_l \cos(kr) \right) = \left(\frac{A_l}{kr} \sin(kr) - \frac{B_l}{kr} \cos(kr) \right) \left(V_0 + \frac{\hbar^2 k^2}{2m} \right)$$

$$-\frac{k}{r} = \frac{1}{kr} \left(V_0 + \frac{\hbar^2 k^2}{2m} \right) \quad -k^2 = V_0 + \frac{\hbar^2 k^2}{2m}$$

1.) b.) cont'd.

$$A_l = \frac{1}{2k} (2l+1) i^l e^{i\delta_l} \quad l=0 \quad \frac{1}{2k} e^{i\delta_0} = A_0$$

$$\tan \delta_l = -\frac{B_l}{A_l} \quad \delta_l = \frac{j_l(ka)}{n_l(ka)}$$

$$j_l(kr) = \frac{1}{kr} \sin(kr)$$

$$B_0 = -A_0 \tan \delta_0$$

for $r < a$

$$B_l = 0$$

$$\psi = A_l j_l(kr)$$

$$\psi = \frac{1}{k^2 r} e^{i\delta_0} \sin(kr)$$

$$\psi'' = -r e^{i\delta_0} \sin(kr)$$

$e^{i\delta_0}$ is just a number
no phase shift in this region $\delta_0 = 0$

$$r \left(-r \sin(kr) + \frac{V_0}{k^2 r} \sin(kr) \right) = \left(\frac{\hbar^2}{2mr} \sin(kr) \right) r$$

$$-r^2 + \frac{V_0}{k^2} = \frac{\hbar^2}{2m}$$

$$r=0 \rightarrow \frac{V_0}{k^2} = \frac{\hbar^2}{2m} \rightarrow$$

$$\psi = \frac{1}{k^2 r} \sin(kr) \quad \text{for } r < a$$

$$k = \sqrt{\frac{2mV_0}{\hbar^2}}$$

$r \geq a$

$$R_{kl} = \frac{\sqrt{A_l^2 + B_l^2}}{kr} \sin(kr + \delta_l)$$

$r < a$

$$\psi'' + V_0 \psi = E \psi$$

$r \geq a$

$$\psi'' = E \psi$$

$$\sqrt{A_0^2 + B_0^2} = \sqrt{A_0^2 + A_0^2 \tan^2 \delta_0}$$

$$\psi = R_{kl} = A_l j_l(kr) + B_l n_l(kr) \quad \tan^2 = \frac{\sin^2}{\cos^2} \quad A_0 \sqrt{1 + \tan^2 \delta_0} \rightarrow \frac{A_0}{\cos \delta_0}$$

$$1 + \frac{\sin^2}{\cos^2} = \frac{1}{\cos^2} (\cos^2 + \sin^2) = \frac{1}{\cos^2}$$

1.) b.) cont'd.

for $r \geq a$
 $\psi =$

$$\frac{A_0}{\cos \delta_0} = \frac{e^{i\delta_0}}{2k \cos \delta_0} \rightarrow \frac{e^{i\delta_0}}{k} \frac{i}{e^{i\delta_0} + e^{-i\delta_0}}$$

$$\cos \delta_0 = \frac{e^{i\delta_0} + e^{-i\delta_0}}{2i}$$

$$e^{-i\delta_0} \left(\cancel{e^{i\delta_0}} + e^{-i\delta_0} \right) \\ 1 + e^{-2i\delta_0}$$

$$\psi = \frac{1}{kr} (1 + e^{-2i\delta_0}) \sin(kr + \delta_0)$$

c.) $\frac{1}{kx} (1 + e^{-2i\delta_0}) \sin(ka + \delta_0) = \frac{1}{k^2 x} \sin(ka)$

$$k(1 + e^{-2i\delta_0}) \sin(\overset{z=ka+\delta_0}{\downarrow} ka + \delta_0) = \sin(ka)$$

$$\sin(z) = \frac{e^{iz} - e^{-iz}}{2i}$$

$$\frac{e^{iz} - e^{-iz}}{2ik} + \frac{e^{-2i\delta_0} e^{ika+i\delta_0} - e^{-2i\delta_0} e^{-ika-i\delta_0}}{2ik} = \frac{e^{ika-i\delta_0} - e^{-ika-3i\delta_0}}{2ik}$$

$$e^{ika+i\delta_0} e^{-i\delta_0} - e^{-ika-i\delta_0} e^{i\delta_0} + e^{ika-i\delta_0} e^{-i\delta_0} - e^{-ika-i3\delta_0} e^{i\delta_0} = \frac{1}{k} \sin ka$$

$$e^{-i\delta_0} e^{ika} \left(e^{2i\delta_0} - e^{-2ika} + 1 - e^{-2ika-2i\delta_0} \right) \\ 1.) c.) \quad e^{2i\delta_0} - e^{-2ika} (1 + e^{-2i\delta_0}) + 1$$

$$\delta_l = \frac{j_l(ka)}{n_l(ka)} = -\tan(ka)$$

$$\delta_0 = -\tan(ka)$$

d.) b/c $k \rightarrow \infty$ as $V_0 \rightarrow \infty$

$\delta_0 \rightarrow -\tan(\infty)$, but I forget what that limit is exactly

$$e.) \sigma_l = \frac{4\pi}{k^2} (2l+1) \sin^2 \delta_l$$

$$\sigma_0 = \frac{4\pi}{k^2} \sin^2(\tan(ka))$$

$$2.) \quad V(r) = \begin{cases} V_0 & r < a \\ 0 & r \geq a \end{cases}$$

$$d\sigma = \frac{m^2}{4\pi^2 \hbar^4} \left| \int V(r) e^{-i\vec{q} \cdot \vec{r}} d^3\vec{r} \right|^2 d\Omega$$

assuming this can be treated like a central potential:

$$f_B(\theta) = \int_0^\infty r^2 dr V(r) \underbrace{\int_0^{2\pi} d\phi}_{2\pi} \int_0^\pi e^{-iqr \cos x} \sin x dx$$

$$u = \cos x, \quad du = -\sin x dx$$

$$\int_{-1}^1 e^{iqr u} du = \frac{1}{iqr} e^{iqr u} \Big|_{-1}^1$$

we pull in the $\frac{1}{4} \rightarrow \frac{1}{2}$
to get $\frac{1}{r} \sin qr$

$$2\pi \int_0^\infty r dr V(r) \sin qr$$

$$2\pi \int_0^a V_0 r \sin qr dr + \int_a^\infty 0 \cdot r \sin qr dr$$

$$f_B(\theta) = 2\pi \int_0^a V_0 r \sin qr dr$$

$$q = 2k \sin \frac{\theta}{2}$$

$$d\sigma = \frac{4m^2}{\hbar^4} \left| \int_0^a V_0 r \sin(qr) dr \right|^2 d\Omega$$

